

Yu Zhi

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INSPIRE Profile: Y.Zhi.3

ACADEMIC CAREER AND EDUCATION

2024–present	PhD Student, Department of Physics, University of Virginia, USA.
2022–2023	M.Phil. in Physics, University of Cambridge, UK.
2018–2022	B.A. in Physics and Mathematics, Colby College, USA. Grade: 3.93/4.0 (Summa Cum Laude).

RESEARCH EXPERIENCE

2024–present	Graduate Research, University of Virginia. Particle Theory/Phenomenology. <i>Advisor: Prof. Julian Heeck.</i>
2022–2023	M.Phil. Research, University of Cambridge. Gravitational Waves and Dark Matter Search with Atom Interferometry. <i>Advisor: Prof. Valerie Gibson.</i>
2021–2022	Honors Thesis, Colby College. Gravity and Symmetry Breaking in Poincaré Gauge Theory. <i>Advisor: Prof. Robert Bluhm.</i>
2020–2021	Research Assistant, Colby College/RPI. Ultracold Neutral Plasma expansion experiments and numerical simulations (NSF-funded).

HONORS, AWARDS, AND FELLOWSHIPS

2022	Honors Degree in Physics, Colby College.
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2022	Sigma Pi Sigma ($\Sigma\Pi\Sigma$) Physics Honor Society.
2018–2022	Dean's List, Colby College.
2021	NSF-funded Research Fellowship (REU).

TEACHING

2024–present	Graduate Teaching Assistant, University of Virginia.
Oct–Dec 2022	Physics Part II Supervisor, University of Cambridge.
May–Sep 2022	Physics Teacher, Illumination Education.
2019–2022	Physics and Math Teaching Assistant, Colby College.

PUBLICATIONS

Summary (04/2026): 5 publications, 40+ citations. Full list:
<https://inspirehep.net/authors/3143930>

5. J. Heeck and Y. Zhi, Gauged Q-balls in flat potentials, arXiv:2604.07454 [hep-ph] (2026).
4. K. Bongs et al. (AION Collaboration), AION-10: Technical Design Report, arXiv:2508.03491 (2025).
3. AION Collaboration, A Prototype Atom Interferometer, arXiv:2504.09158 (2025).
2. R. Bluhm and Y. Zhi, Spontaneous and Explicit Spacetime Symmetry Breaking, Symmetry 16 (2024) 1, 25.
1. B. Stray et al., Centralized design and production for AION, AVS Quantum Sci. 6 (2023) 1, 014409.

SELECTED TALKS

Aug 2023	MAGIS Collaboration Conference, Northwestern University, USA.
Apr 2023	IOP Joint APP and HEPP Annual Conference, King's College London, UK.
Apr 2022	Physics and Astronomy Honors Colloquium, Colby College, USA.